

# Transit or Mirage: Machine Learning Classification of Kepler Planet Candidates

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## 1 Introduction

Launched in 2009, the Kepler Space Telescope was NASA’s first mission dedicated to finding planets outside our solar system. Monitoring over 150,000 stars in the constellation Cygnus, Kepler confirmed more than 2,600 exoplanets before its retirement in October 2018 ([NASA, 2026](#)).

Kepler detects planets through the transit method: when a planet passes directly between its host star and the telescope, it blocks a fraction of the star’s light, producing a measurable dip in the star’s light curve, calculated by

$$\delta = \left( \frac{R_p}{R_*} \right)^2 \quad (1)$$

where  $R_p$  is the planetary radius and  $R_*$  is the stellar radius ([University of Nebraska-Lincoln, 2024](#)).

Despite Kepler’s success, not every transit-like signal corresponds to a real planet. Astrophysical false positives (signals that mimic planetary transits but originate from other phenomena) frequently contaminate the candidate catalog. The most common sources are eclipsing binaries and their variants (grazing and background), which produce periodic brightness dips or diluted eclipse signals that mimic planetary transits. Across the full KOI catalog, the global false positive rate has been estimated at approximately 9.4%, though the rate varies significantly with planet size and orbital period ([Fressin et al., 2013](#)). Furthermore, misclassification carries real costs in both directions: a missed planet may remove a candidate from consideration, while a false positive wastes expensive radial velocity follow-up time that could otherwise be spent confirming real discoveries.

Missing a real planet (a false negative) is a worse outcome, however, and the asymmetry in error costs motivates prioritizing false negative reduction over false positive reduction. This project trains and compares two ensemble classifiers, Random Forest and XGBoost, on the NASA Kepler Objects of Interest cumulative table to distinguish confirmed planets from false positives. Model predictions are interpreted using SHAP values in Section 5.4 to connect learned feature importance back to the underlying astrophysics, and both models are benchmarked against NASA’s own disposition scores in Section 6. The trained classifiers are then applied to 1,979 unresolved candidates to produce

a prioritized target list for radial velocity follow-up in Section 6.3, connecting directly to active research efforts extracting exoplanet signals from RV time series.

## 2 Data

### 2.1 The KOI Cumulative Table

The dataset used in this project is the Kepler Objects of Interest (KOI) cumulative table, obtained from the NASA Exoplanet Archive ([NASA Exoplanet Science Institute, 2026](#)). Each row corresponds to a transit signal detected by Kepler, labeled as **CONFIRMED** (2,746 planets), **FALSE POSITIVE** (4,839 signals), or **CANDIDATE** (1,979 unresolved). The confirmed and false positive rows formed the labeled training set, while candidates were set aside for final prediction.

### 2.2 Feature Selection and Engineering

Features were selected based on physical relevance to transit classification. Kept features include transit geometry parameters (`koi_depth`, `koi_ror`, `koi_impact`), planet and orbital properties (`koi_prad`, `koi_period`, `koi_dor`), signal quality metrics (`koi_model_snr`, `koi_max_mult_ev`), and host star properties. Identifier columns and time-based metadata with no physical bearing on transit classification were dropped. Columns with no data were removed.

The column `koi_score` is NASA’s precomputed disposition confidence score, which is derived from the same vetting process that produced the class labels. Including it would inflate model accuracy without the model meaningfully learning, so it was excluded.

### 2.3 Class Imbalance

When initially exploring the data, it was noted that there were substantially more **FALSE POSITIVE** labels than there were **CONFIRMED**, with a ratio of 4,839 to 2,746 (64/36). This imbalance is addressed in the modeling pipeline, discussed in Section 4.

## 3 Exploratory Data Analysis

All figures below are plotted on a logarithmic scale, due to a consistent heavy right skew.

### 3.1 Orbital Period (koi\_period)

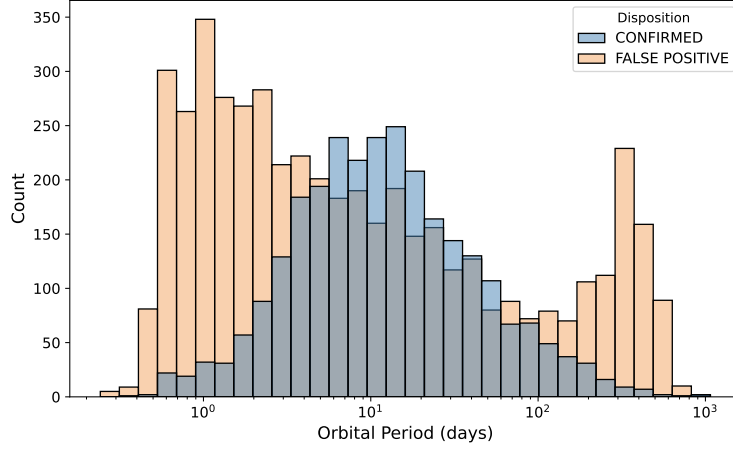


Figure 1: Distribution of orbital period by disposition plotted on a log scale. False positives concentrate at very short periods and show an unexpected spike around 300-500 days.

There is significant overlap between **CONFIRMED** and **FALSE POSITIVE** labels for **koi\_period**; however, the concentrations of false positives at orbital periods of approximately 1 day and another spike between periods of 100 to 1,000 days are noteworthy observations.

### 3.2 Model SNR (koi\_model\_snr)

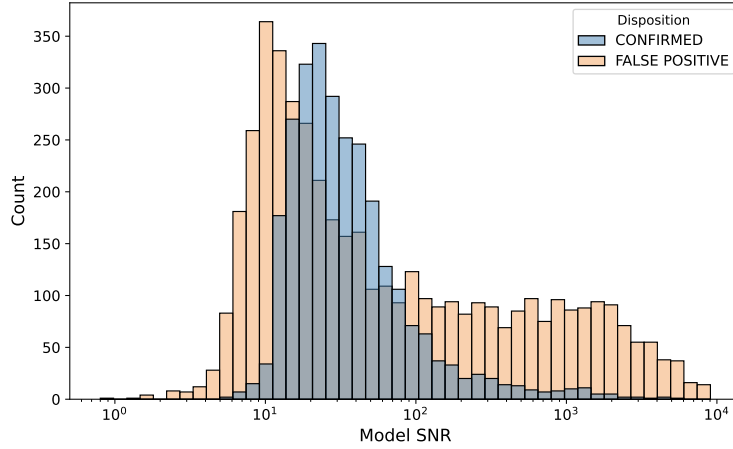


Figure 2: Distribution of model SNR by disposition plotted on a log scale. False positives dominate at high SNR, which is counterintuitive at first, but it could be explained by eclipsing binaries producing extremely large and clean signals.

This distribution represents the signal-to-noise ratio, which measures the planetary transit signal's strength against background noise. A higher SNR means a clear detection, whereas a lower bound SNR of roughly 7.1 is commonly employed to differentiate between real and fake transit candidates (Kunimoto et al., 2018). The data show the lowest model SNR of a planet labeled **CONFIRMED** is 5.9.

### 3.3 Impact Parameter (`koi_impact`)

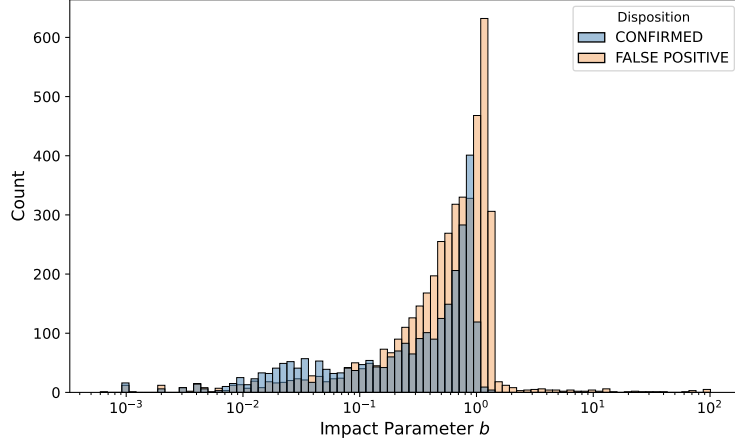


Figure 3: Distribution of impact parameter  $b$  by disposition plotted on a log scale. A full transit requires  $b < 1$ , so false positives with  $b \gg 1$  are likely grazing eclipsing binaries rather than real transits.

The impact parameter  $b$  parameterizes the sky-projected distance between the transit chord and the stellar center in units of the stellar radius. 1,198 values labeled **FALSE POSITIVE** have  $b > 1$ , ranging from 1.001 to 100.806 and an average of roughly 2.214. In contrast, only 16 values labeled **CONFIRMED** have a  $b > 1$ , and they range from 1.004 to 1.3, with an average of roughly 1.19. Therefore,  $b > 1$  values end up being a strong predictor of a **FALSE POSITIVE**.

### 3.4 Radius Ratio (`koi_ror`)

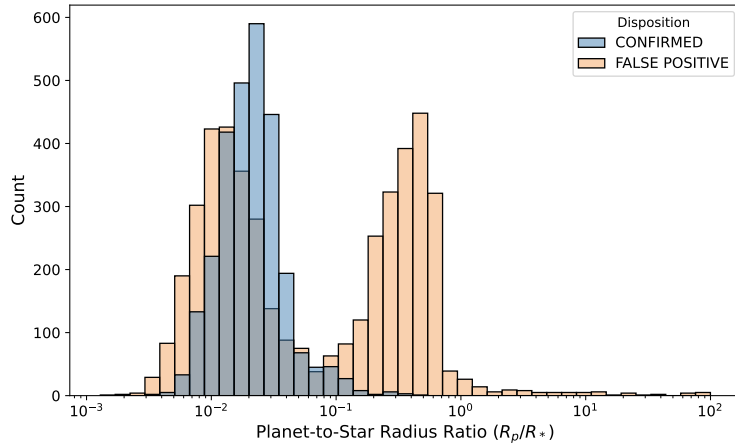


Figure 4: Distribution of planet-to-star radius ratio by disposition plotted on a log scale. The false positive distribution is bimodal, with a low-depth population overlapping with real planets and a high-depth population of false positives.

The planet-to-star radius ratio contains a clean bimodal distribution, with **CONFIRMED** and **FALSE POSITIVE** labels being cleanly distinguishable. Since  $\delta = (R_p/R_*)^2$  and due to their mathematical relationship, `koi_ror` is representative of two other features in the dataset: `koi_depth` and `koi_prad`.

$koi\_ror = R_p/R_*$ ,  $koi\_depth = koi\_ror^2$ , and  $koi\_prad$  is simply  $koi\_ror$  scaled by the known stellar radius.

## 4 Methods

The full code and analysis are available at <https://github.com/Swaraag/ExoplanetClassifier>.

### 4.1 Pipeline

I split the data 80/20 into train and test sets and chained imputation and modeling inside a `scikit-learn Pipeline` (Pedregosa et al., 2011) to prevent data leakage from the test set into the trained model. Two ensemble tree models were tested: a Random Forest Classifier and an XGBoost Classifier. Both handle tabular data well without scaling and are robust to skewed distributions.

Removing any of the three highly correlated variables discussed in Section 3.4 risks eliminating useful split points deeper in the tree, since each feature encodes the same physical information at slightly different scales that individual trees can exploit. Therefore, I included all of them.

### 4.2 Imputation

Table 1: Features with missing values, grouped by count.

| Missing | Features                                                                                                                                                                                                                                                                                                   |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 910     | <code>koi_bin_oedp_sig</code>                                                                                                                                                                                                                                                                              |
| 681     | <code>koi_num_transits</code> , <code>koi_max_mult_ev</code> , <code>koi_max_sngle_ev</code>                                                                                                                                                                                                               |
| 279     | <code>koi_smet</code>                                                                                                                                                                                                                                                                                      |
| 260     | <code>koi_incl</code>                                                                                                                                                                                                                                                                                      |
| 259     | <code>koi_model_snr</code> , <code>koi_dor</code> , <code>koi_steff</code> , <code>koi_teq</code> , <code>koi_smass</code> ,<br><code>koi_prad</code> , <code>koi_slogg</code> , <code>koi_ror</code> , <code>koi_depth</code> , <code>koi_impact</code> ,<br><code>koi_srad</code> , <code>koi_sma</code> |
| 219     | <code>koi_insol</code> , <code>koi_srho</code>                                                                                                                                                                                                                                                             |

Due to the heavy right-skew of data, I chose to employ a median imputation to deal with the missing values described in Table 1, enabled by a `scikit-learn SimpleImputer`. Median imputation is more robust to extreme values, making it the better choice for this dataset. Features with the highest number of missing values, particularly `koi_bin_oedp_sig`, which was missing in roughly 12% of the dataset, may carry imputation-induced bias. More sophisticated imputation approaches could reduce this bias, but were not employed here for simplicity.

### 4.3 Random Forest Classifier

Random Forest is a bagging-based ensemble of decision trees that reduces variance through bootstrap sampling and random feature selection at each split. The classifier used 100 trees with

`class_weight=balanced`, which weights classes inversely proportional to their frequencies to address the class imbalance discussed in Section 2.3.

#### 4.4 XGBoost Classifier

XGBoost is a gradient boosted tree ensemble in which trees are built sequentially to correct residual errors, with a regularization term to penalize complexity (Chen and Guestrin, 2016). The classifier used 100 trees with `scale_pos_weight` set to  $n_{\text{negative}}/n_{\text{positive}}$  to address class imbalance, as XGBoost lacks an equivalent `class_weight=balanced` parameter.

#### 4.5 Evaluation Strategy

Accuracy alone is insufficient for these purposes: a model that always predicts FALSE POSITIVE would achieve 64% accuracy without learning anything due to the class imbalance. I therefore evaluate using ROC-AUC, confusion matrices (with particular attention to false negatives, given the asymmetric error costs discussed in Section 1), and 5-fold stratified cross-validation that preserves the 64/36 class ratio in each fold.

### 5 Results

#### 5.1 Model Performance

Table 2: Performance comparison of Random Forest and XGBoost classifiers. CV AUC is the mean  $\pm$  standard deviation over 5 stratified folds.

| Model         | Accuracy | ROC-AUC | CV AUC             | False Negatives |
|---------------|----------|---------|--------------------|-----------------|
| Random Forest | 92.7%    | 0.980   | $0.9780 \pm 0.015$ | 53              |
| XGBoost       | 94.5%    | 0.986   | $0.9807 \pm 0.014$ | 28              |

The XGBoost classifier outperformed Random Forest across every metric, and the very tight cross-validation standard deviations ( $\pm 0.015$  and  $\pm 0.014$ ) confirm that neither model is overfitting to a lucky split.

#### 5.2 Confusion Matrix

Table 3: Confusion matrices for Random Forest and XGBoost. Rows are actual labels, while columns are the predicted labels.

| Actual         | Random Forest |              | XGBoost  |              |
|----------------|---------------|--------------|----------|--------------|
|                | Pred. FP      | Pred. Planet | Pred. FP | Pred. Planet |
| False Positive | 902           | 58           | 904      | 56           |
| Confirmed      | 53            | 504          | 28       | 529          |

Table 3 shows a clear asymmetry in XGBoost’s false positive and false negative rates. As a fraction of confirmed planets in the test set, XGBoost’s false negative rate was  $\frac{28}{557} \approx 5.0\%$  compared to Random Forest’s  $\frac{53}{557} \approx 9.5\%$ , nearly halving the rate of missed planets.

### 5.3 Feature Importances

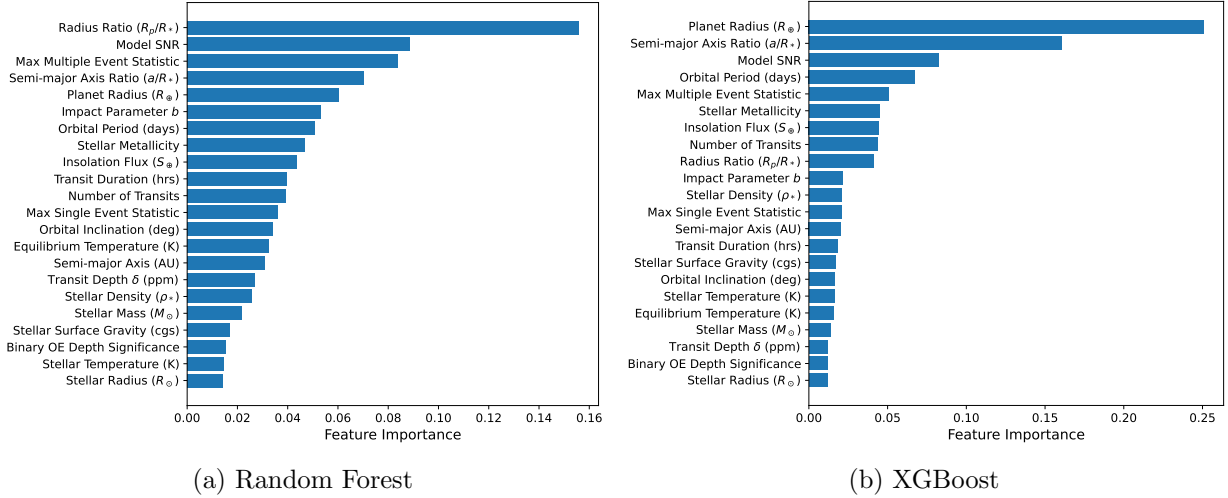


Figure 5: Feature importances for both models. Radius Ratio ( $R_p/R_*$ ) dominates in Random Forest, while Planet Radius ( $R_p$ ) tops XGBoost, despite both features encoding essentially the same physical information.

`koi_ror` dominates the Random Forest importance ranking because a single threshold on it already cleanly separates most of the data, maximizing Gini impurity reduction at the top of nearly every tree. `koi_prad` and `koi_depth` rank lower despite being physically informative, as their high correlation with `koi_ror` leaves little additional impurity to reduce once it has already split.

Surprisingly, `koi_prad` lies at the top of the XGBoost feature importance graph. As discussed in Section 3.4, `koi_prad` is simply `koi_ror` scaled by stellar radius, so this additional variation likely gives XGBoost’s sequential trees more to exploit, inflating its gain-based importance relative to `koi_ror`.

### 5.4 SHAP Analysis

Inspired by the XGBoost discrepancy discussed in Section 5.3, I used the Python library `shap` to investigate further (Lundberg and Lee, 2017). SHAP (SHapley Additive exPlanations) measures the contribution of each feature to individual predictions, thereby providing a much greater level of interpretability on XGBoost’s predictions.

Figure 6 offers clues into XGBoost’s approach: the plot describes, for every feature, how much it pushed each individual predictions towards or away from being an exoplanet. Blue values correspond to lower variable values, whereas red values correspond to the higher ones.

### 5.4.1 SHAP Summary

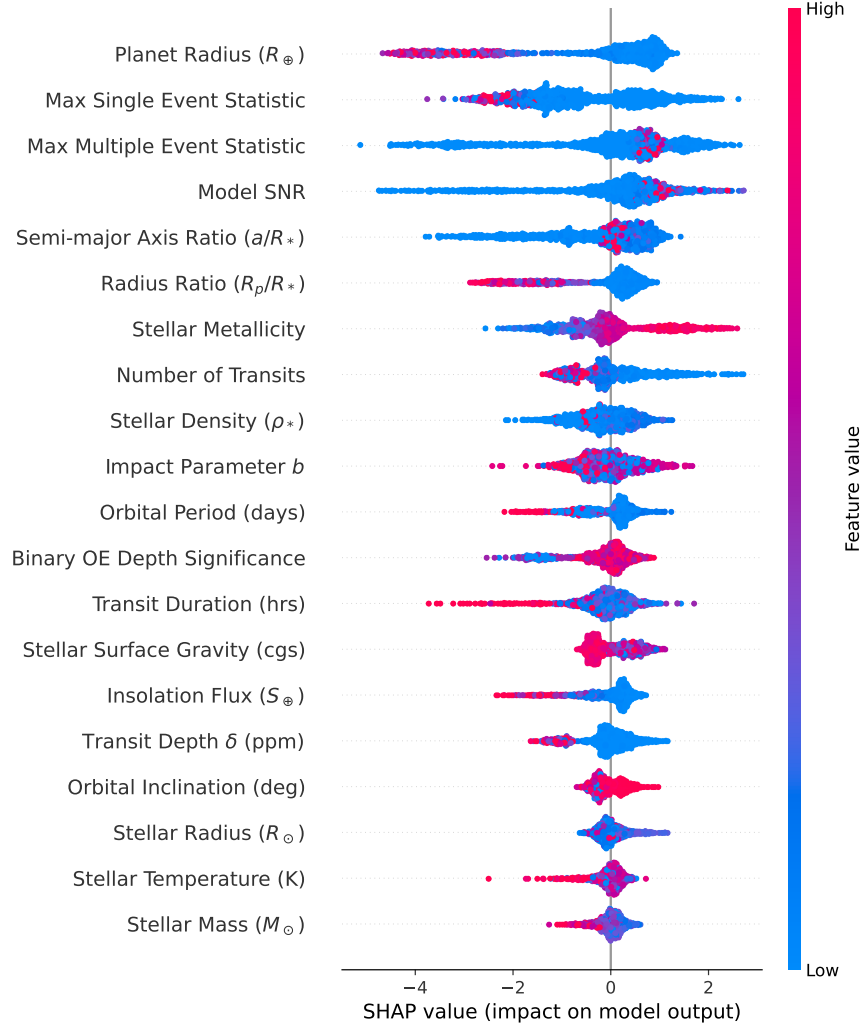


Figure 6: SHAP summary plot for XGBoost. Each point represents one test sample, colored by feature value (red = high, blue = low). Planet Radius dominates, where low values push strongly toward a planet prediction, and high values push toward false positives.

Features near the bottom of the plot, such as the stellar-related variables, have a narrower spread of datapoints. The tight clustering near 0 indicates the features barely move predictions in either direction, which is expected for stellar variables. I highlight three of the more interesting variables below.

Planet Radius ( $R_{\oplus}$ ) / `koi_prad`: the samples where the planet radius is small (blue) have large positive SHAP values, meaning a small radius pushed the model strongly toward predicting an exoplanet. The samples where the planet radius is large (red) have large negative SHAP values, meaning a large radius pushed the model strongly toward **FALSE POSITIVE**.

Model SNR / `koi_model_snr`: the red dots (a high SNR) push left toward **FALSE POSITIVE**, whereas the blue dots (low SNR) push toward **CONFIRMED**. This was a feature noted in Figure 2



to be counterintuitive at first, because one might expect a cleaner signal to mean a more reliable planet detection.

Stellar Metallicity / `koi_smet`: stellar metallicity ranks higher than expected for a feature that does not directly describe a transit signal. The model seems to have learned a curious population-level trend that might explain this variable’s impact. Host star metallicity correlates positively with planet occurrence, with the fraction of stars hosting gas giant planets rising from under 3% at subsolar metallicity to 25% at  $[\text{Fe}/\text{H}] > +0.3$  dex (Fischer and Valenti, 2005).

#### 5.4.2 SHAP Waterfall

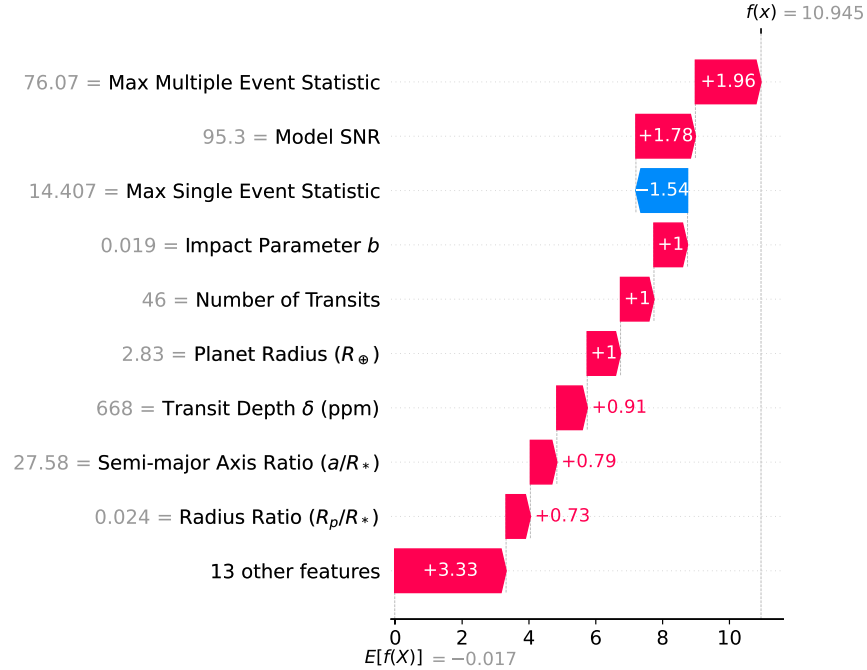


Figure 7: SHAP waterfall plot for the most confidently predicted planet in the test set ( $R_p = 2.83 R_{\oplus}$ ). Starting from the base value  $E[f(X)] = -0.017$ , nearly every feature pushes toward predicting **CONFIRMED**, with Max Multiple Event Statistic (+1.96), Model SNR (+1.78), and 13 other features (+3.33) driving the final output to  $f(x) = 10.945$ . Max Single Event Statistic is the only notable feature pushing the model toward a **FALSE POSITIVE** (−1.54).

Figure 7 indicates how much each feature individually contributes to the prediction made by the model on one specific datapoint. This type of plot is essential for determining causes for false positives and negatives by exploring which variables enabled the incorrect prediction.

## 6 NASA Comparison and Candidate Predictions

### 6.1 Comparison with NASA Disposition Scores

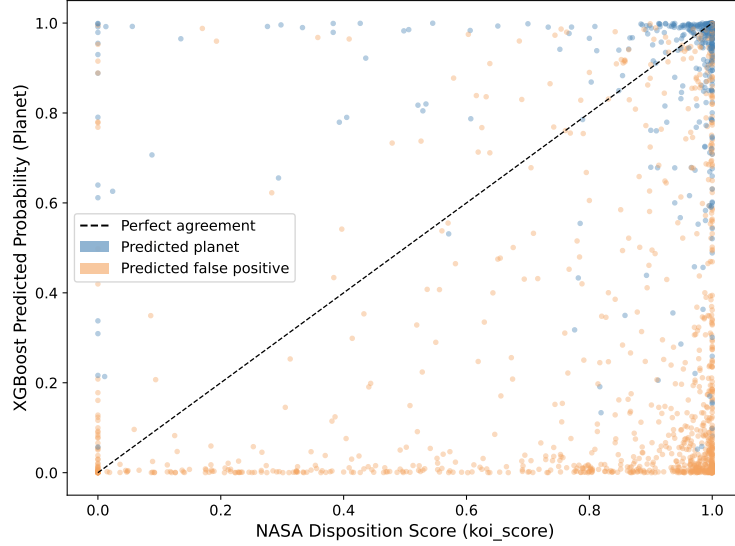


Figure 8: XGBoost predicted planet probability vs. NASA disposition score for all 1,979 unresolved candidates. Agreement is strong at both extremes, with meaningful scatter in the middle where ambiguous candidates live.

Candidates in the bottom-right quadrant (low model probability, high NASA score) are the most interesting disagreements in Figure 8.

### 6.2 Error Distribution Across NASA Disposition Scores

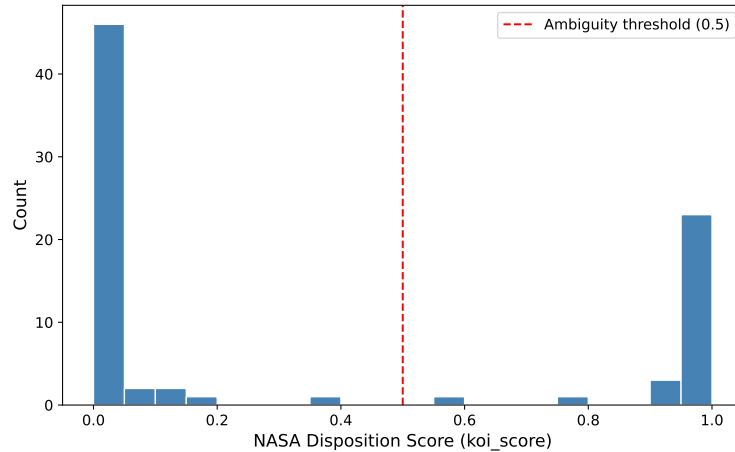


Figure 9: NASA disposition scores for test set samples XGBoost classified incorrectly. The heavy concentration near a score of 0 (46 errors) vs. a score of 1 (23 errors) shows that XGBoost is twice as likely to flag false positives as planets than it is to miss a real planet.

The smaller peak near a score of 1.0 in Figure 9 is the more important result, since they are real planets that XGBoost missed. The larger peak near 0.0, while representing more total errors, consists of false positives incorrectly flagged as planets, which is a less costly mistake since follow-up observations could quickly rule them out.

### 6.3 Candidate Predictions

Of the 1,979 unresolved candidates run through XGBoost, 394 were flagged as likely planets. Of these 394, 152 were in the sub-Neptune range ( $1.7 R_{\oplus}$ - $3.5 R_{\oplus}$ ; Hu et al. 2021), peaking around 2.0-2.1  $R_{\oplus}$ . The size distribution is consistent with known planetary occurrence rates (Fressin et al., 2013), suggesting the model is producing physically realistic predictions rather than spurious ones. These 152 candidates are particularly valuable because sub-Neptunes sit at the boundary between rocky super-Earths and gas-dominated planets (Fulton et al., 2017), making their atmospheric compositions poorly constrained.

Radial velocity (RV) measurements, which detect the gravitational wobble a planet induces on its host star, remain the standard method for confirming transit candidates and measuring their masses, so radial velocity follow-up of these candidates would directly contribute to confirming or ruling out their planetary nature (Beard et al., 2025).

Of the 394 candidates flagged as likely planets, 320 (81%) have NASA disposition scores above 0.90, and 349 (89%) exceed 0.75, indicating strong agreement between XGBoost’s model and NASA’s own vetting pipeline. The remaining 34 candidates with NASA scores below 0.5 represent cases where the model and NASA disagree, warranting closer inspection.

## 7 Conclusion

This project trained and compared Random Forest and XGBoost classifiers on the Kepler Objects of Interest catalog to automate exoplanet candidate vetting. XGBoost outperformed Random Forest across every metric, achieving a ROC-AUC of 0.986 and notably fewer missed planets (28 vs. 53 false negatives). SHAP analysis confirmed that predictions are driven by physically meaningful features, and the model identified 394 high-confidence planet candidates from 1,979 unresolved KOIs.

A SHAP waterfall on the 34 candidates where XGBoost and NASA disagree (scores below 0.5) is an actionable next step to identify which features are driving the divergence. Future work could also extend this approach to TESS data using the same feature structure, or replace the tabular summary statistics with raw light curve data to capture what the current pipeline misses.

Additionally, the current pipeline uses default hyperparameters and a fixed 0.5 decision threshold; tuning both (particularly lowering the threshold to reflect the higher cost of missed planets) could further reduce false negatives. Evaluating with precision-recall metrics and testing sensitivity to median imputation on heavily missing features like `koi_bin_oedp_sig` would also strengthen

confidence in the results.

## References

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